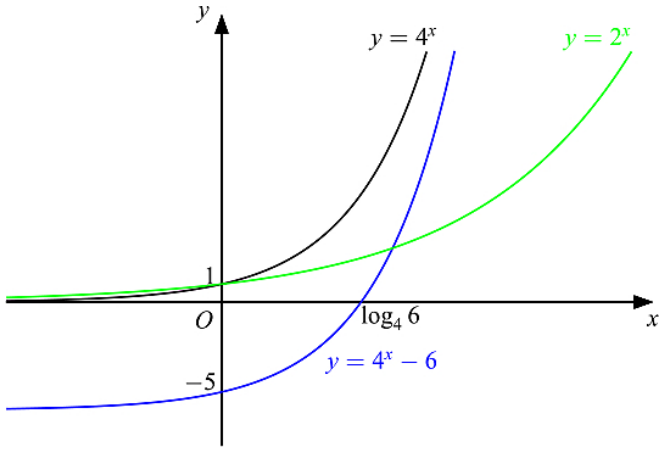
1(a)	(i) There is a common difference (no common ratio) and so an arithmetic series should be used.	**B1**
	(ii) $(u_n) = 5 + (n-1)"d"$ or $(u_n) = 5 + n"d"$	**M1**
	$(u_n) = 5 + 0.25(n-1)$ oe	**A1**
		(3)
(b)	Need $S_n = \frac{n}{2}\left(2\times 5 + (n-1)\times 0.25\right) \geqslant 350$	**M1A1**
	$\Rightarrow 0.25n^2 + 9.75n \geqslant 700 \Rightarrow (n+19.5)^2 - 19.5^2 \geqslant 2800 \Rightarrow n = ...$	**M1**
	So 37 weeks.	**A1**
		(4)

2(a)		(i) Correct shape and position for $y = 2^x$ Crosses y axis with same intersection as $y = 4^x$ but with gentler slope.	**B1**
		(ii) Graph of same shape as $y = 4^x$ but translated down	**B1**
		y intercept at -5	**B1**
		x intercept at $\log_4 6$	**B1**
			(4)
(b)	$2^x = 4^x - 6$ or $y = (2^x)^2 - 6 = y^2 - 6$		**M1**
	$\Rightarrow 2^{2x} - 2^x - 6 = 0 \Rightarrow (2^x - 3)(2^x + 2) = 0 \Rightarrow 2^x = \ldots$ or $(y-3)(y+2) = 0 \Rightarrow y = \ldots$		**M1**
	$y = 2^x = 3$		**A1**
	$x = \log_2 3$ oe		**A1**
			(4)

3(i)	At least three of: For $p=2:\ 2^3+2=8+2=10$; For $p=3:\ 3^3+3=27+3=30$ For $p=5:\ 5^3+5=125+5=130$; For $p=7:\ 7^3+7=343+7=350$	**M1**
	Each case gives a multiple of 10. As 2,3,5 and 7 are the only single digit primes, the result has been proved for all single digit primes.	**A1**
		(2)
(ii)	$(n+1)^3-n^3=n^3+3n^2+3n+1-n^3=3n^2+3n+1$	**M1A1**
	$=3(n^2+n)+1$ which is one more than a multiple of 3, so is not divisible by 3 for any $n\in\mathbb{N}$	**A1**
		(3)

4(a)	$\left(2+\frac{x}{8}\right)^{13} = 8192 + \ldots$	**B1**
	$+\underline{\binom{13}{1}}2^{12}\left(\frac{x}{8}\right)^{1} + \underline{\binom{13}{2}}2^{11}\left(\frac{x}{8}\right)^{2} + \underline{\binom{13}{3}}2^{10}\left(\frac{x}{8}\right)^{3} + \ldots$	**M1**
	$\left(2+\frac{x}{8}\right)^{13} = (8192) + 6656x + 2496x^2 + 572x^3 + \ldots$	**A1A1**
		(4)
(b)	$\frac{x}{8} = 0.0125 \Rightarrow x = 0.1$	**B1**
	$2.0125^{13} \approx 8192 + 665.6 + 24.96 + 0.572$	**M1**
	$= 8883.132$ cao	**A1**
		(3)
(c)	As all the terms in the expansion are positive, the truncated series will give an underestimate of the actual value.	**B1**
		(1)

5(a)	$h = \frac{1}{4}$	**B1**
	$A \approx \left(\frac{1}{2} \times \frac{1}{4}\right)\left\{5.453 + 5 + 2\left(7.764 + ... + 7.626\right)\right\}$	**M1**
	$A \approx \left(\frac{1}{2} \times \frac{1}{4}\right)\left\{5.453 + 5 + 2\left(7.764 + 9.375 + 9.964 + 9.367 + 7.626\right)\right\}$	**A1**
	= awrt 12.33	**A1**
		(4)
(b)	$\int x^{\frac{3}{2}} - 3x + 9 \, \mathrm{d}x = kx^{\frac{5}{2}} - lx^2 + mx$ (at least two powers correct)	**M1**
	$\int x^{\frac{3}{2}} - 3x + 9 \, \mathrm{d}x = \frac{2}{5}x^{\frac{5}{2}} - \frac{3}{2}x^2 + 9x \, (+c)$	**A1A1**
		(3)
(c)	$\left[\frac{2}{5}x^{\frac{5}{2}} - \frac{3}{2}x^2 + 9x\right]_{\frac{5}{2}}^{4} = \left(\frac{2}{5} \times 4^{\frac{5}{2}} - \frac{3}{2} \times 4^2 + 9 \times 4\right) - \left(\frac{2}{5}\left(\frac{5}{2}\right)^{\frac{5}{2}} - \frac{3}{2}\left(\frac{5}{2}\right)^2 + 9\left(\frac{5}{2}\right)\right) = ...$ $\left(= 24.8 - 17.077... = 7.722...\right)$	**M1**
	Area R = "12.33" – "7.722" = …	**M1**
	= awrt 4.6	**A1**
		(3)

6(a)	Way 1: Eqn is $(x-3)^2+(y+4)^2-9-16+k=0$	**M1**
	So radius must be 4 $\Rightarrow 25-k=16$	**M1**
	$\Rightarrow k=9$	**A1**
		(3)
	Way 2: $y=0 \Rightarrow x^2-6x+k=0$ has one solution	**M1**
	$\Rightarrow 6^2-4\times1\times k=0$	**M1**
	$\Rightarrow k=9$	**A1**
		(3)
(b)	C intersects x axis at (3,0)	**B1**
	Intersects y axis when $y^2+8y+9=0 \Rightarrow y=...$ or $y="-4"\pm\sqrt{"16"-"9"}$ Or uses base of triangle is $2\sqrt{"16"-"9"}$	**M1**
	Area $RST=\frac{1}{2}\times2\sqrt{7}\times3$	**M1**
	$=3\sqrt{7}$	**A1**
		(4)

7(a)	$2 = \log_3 3^2$ oe seen or implied by working	**B1**
	$3\log_3(2x-1) = \log_3(2x-1)^3$ or $\log_3"3^2" + \log_3(14x-25) = \log_3\left("3^2"(14x-25)\right)$	**M1**
	$\Rightarrow (2x-1)^3 = 9(14x-25)$	**dM1**
	$\Rightarrow 8x^3 - 12x^2 + 6x - 1 = 126x - 225$ $\Rightarrow 8x^3 - 12x^2 - 120x + 224 = 0 \Rightarrow 2x^3 - 3x^2 - 30x + 56 = 0$*	**A1***
		(4)
(b)	$2(\pm 4)^3 - 3(\pm 4)^2 - 30(\pm 4) + 56 = ...$	**M1**
	$2(-4)^3 - 3(-4)^2 - 30(-4) + 56 = -128 - 48 + 120 + 56 = 0$ Hence −4 is a root of the equation.	**A1**
		(2)
Alt (b)	$2x^3 - 3x^2 - 30x + 56 = (x \pm 4)\left(2x^2 + ... \pm 14\right)$	**M1**
	$2x^3 - 3x^2 - 30x + 56 = (x+4)\left(2x^2 - 11x + 14\right)$ so −4 is a root of the equation.	**A1**
		(2)
(c)	$2x^3 - 3x^2 - 30x + 56 = 0 \Rightarrow (x+4)\left(2x^2 + ... \pm 14\right) = 0$	**M1**
	$(x+4)\left(\underline{2x^2 - 11x + 14}\right) = 0$	**A1**
	$(x+4)(2x-7)(x-2) = 0 \Rightarrow x = ...$	**dM1**
	(Equation not defined for $x = -4$ so) solutions are 2 and $\frac{7}{2}$	**A1**
		(4)

8(i)	$3\sin(\theta+30°)=7\cos(\theta+30°)\Rightarrow\tan(\theta+30°)=\frac{7}{3}$	**B1**
	$\theta+30°=\arctan\left(\frac{7}{3}\right)\ (=66.8°)$	**M1**
	$\theta=36.8°$ or $216.8°$ (awrt)	**A1A1**
		(4)
(ii)	(a) $3\sin^3 x=5\sin x-7\sin x\cos x\Rightarrow 3\sin x\left(1-\cos^2 x\right)=5\sin x-7\sin x\cos x$	**M1**
	$\Rightarrow\sin x\left(5-7\cos x+3\cos^2 x-3\right)=0$	
	$\Rightarrow\sin x\left(3\cos^2 x-7\cos x+2\right)=0$	**A1**
	(b) $\sin x=0\Rightarrow x=0$	**B1**
	$\Rightarrow\sin x\left(3\cos x-1\right)\left(\cos x-2\right)=0\Rightarrow\cos x=...$	**M1**
	$3\cos x=1\Rightarrow x=\arccos\left(\frac{1}{3}\right)=(\pm)1.23$	**M1**
	Both of x = awrt -1.23, 1.23	**A1**
		(6)

	Alternatives to (i)	
8(i) Alt 1	$\sin(\theta+30°)=\frac{7}{\sqrt{58}}$ or $\cos(\theta+30°)=\frac{3}{\sqrt{58}}$	**B1**
	$(3\sin(\theta+30°))^2=7(\cos(\theta+30°))^2 \Rightarrow ..\sin^2(\theta+30°)=..(1-\sin^2(\theta+30°))$ $\Rightarrow \sin(\theta+30°)=..\Rightarrow \theta+30°=...$ Or $(3\sin(\theta+30°))^2=7(\cos(\theta+30°))^2 \Rightarrow ..(1-\cos^2(\theta+30°)=..\cos^2(\theta+30°)$ $\Rightarrow \cos(\theta+30°)=..\Rightarrow \theta+30°=...$	**M1**
	$\theta=36.8°$ or $216.8°$ (awrt)	**A1A1**
		(4)
8(i) Alt 2	$3\sin(\theta+30°)=7\cos(\theta+30°)=\frac{3\sqrt{3}}{2}\sin\theta+\frac{3}{2}\cos\theta=\frac{7\sqrt{3}}{2}\cos\theta-\frac{7}{2}\sin\theta$ oe	**B1**
	$3\sin(\theta+30°)=7\cos(\theta+30°)\Rightarrow$ $3(\sin\theta\cos30°\pm\cos\theta\sin30°)=7(\cos\theta\sin30°\pm\sin\theta\sin30°)$ $\Rightarrow \frac{\sin\theta}{\cos\theta}=\tan\theta=...\Rightarrow\theta=\arctan(..)=...$	**M1**
	$\theta=36.8°$ or $216.8°$ (awrt)	**A1A1**
		(4)

9(a)	$V = hl^2 \Rightarrow 250000 = hl^2$ or $l = \dfrac{500}{\sqrt{h}}$ oe (may use e.g. $hl = \sqrt{250000h}$)	**B1**
	$S = l^2 + 4hl = "\dfrac{250000}{h}" + 4h \times "\dfrac{500}{\sqrt{h}}"$	**M1**
	$S = \dfrac{250000}{h} + 2000\sqrt{h}$ *	**A1***
		(3)
(b)	$\dfrac{\mathrm{d}S}{\mathrm{d}h} = -\dfrac{250000}{h^2} + 2000 \times \dfrac{1}{2} h^{-\frac{1}{2}}$ oe	**M1A1**
	$\dfrac{\mathrm{d}S}{\mathrm{d}h} = 0 \Rightarrow -\dfrac{250000}{h^2} + 2000 \times \dfrac{1}{2} h^{-\frac{1}{2}} = 0 \Rightarrow h^k = ...$	**dM1**
	$\Rightarrow h^{\frac{3}{2}} = 250 \Rightarrow h = ...$	**ddM1**
	$h = 250^{\frac{2}{3}}$	**A1**
		(5)
(c)	$\dfrac{\mathrm{d}^2S}{\mathrm{d}h^2} = \dfrac{500000}{h^3} - 500h^{-\frac{3}{2}}$	**M1**
	$\left.\dfrac{\mathrm{d}^2S}{\mathrm{d}h^2}\right\|_{h=39.7} = 6 > 0$ hence gives the minimum value.	**A1**
		(2)